Method of expressing concentration of solution

1. $25ml$ of $3.0MHNO_3$ are mixed with $75ml$ of $4.0MHNO_3$. If the volumes are additive, the molarity of the final mixture would be
(a) $3.25M$ (b) $4.0M$
(c) $3.75M$ (d) $3.50M$

2. The amount of anhydrous Na_2CO_3 present in 250 *ml* of 0.25 *M* solution is
(a) 6.225 *g* (b) 66.25 *g*
(c) 6.0 *g* (d) 6.625 *g*

3. Dilute one litre 1 molar H_2SO_4 solution by 5 litre water, the normality of that solution is
(a) $0.2N$ (b) $5N$
(c) $10N$ (d) 0.33 *N*

4. If 5.85 gms of $NaCl$ are dissolved in 90 gms of water, the mole fraction of $NaCl$is
(a) 0.1 (b) 0.2
(c) 0.3 (d) 0.01
(e) 0.0196

5. The molarity of 0.006 mole of $NaCl$ in $100ml$ solution is
(a) 0.6 (b) 0.06
(c) 0.006 (d) 0.066
(e) None of these

6. $9.8g$ of H_2SO_4 is present in 2 *litres* of a solution. The molarity of the solution is
(a) $0.1M$ (b) $0.05M$
(c) $0.2M$ (d) $0.01M$

7. What will be the molarity of a solution containing $5g$ of sodium hydroxide in $250ml$ solution
(a) 0.5 (b) 1.0
(c) 2.0 (d) 0.1

8. The normality of $0.3M$ phosphorus acid (H_3PO_3) is
(a) 0.1 (b) 0.9
(c) 0.3 (d) 0.6

9. Which of the following has maximum number of molecules
(a) 16 *gm* of O_2
(b) 16 *gm* of NO_2
(c) 7 *gm* of N_2
(d) 2 *gm* of H_2

10. Molarity is expressed as
(a) Gram/litre
b) Moles/litre
(c) Litre/mole
d) Moles/1000 *gms*

11. $20ml$ of HCl solution requires $19.85ml$ of $0.01MNaOH$ solution for

complete neutralization. The molarity of HCl solution is

(a) 0.0099 (b) 0.099
(c) 0.99 (d) 9.9

12. How much of *NaOH* is required to neutralise 1500 cm^3 of 0.1 *N HCl* (At. *wt*. of *Na* =23)

(a) 4 *g* (b) 6 *g*
(c) 40 *g* (d) 60 *g*

13. If 5.85 *g* of *NaCl* (molecular weight 58.5) is dissolved in water and the solution is made up to 0.5 *litre*, the molarity of the solution will be

(a) 0.2 (b) 0.4
(c) 1.0 (d) 0.1

14. A mixture has 18*g* water and 414*g* ethanol. The mole fraction of water in mixture is (assume ideal behaviour of the mixture)

(a) 0.1 (b) 0.4
(c) 0.7 (d) 0.9

15. The number of molecules in 4.25 *g* of ammonia is approximately

(a) 0.5×10^{23} (b) 1.5×10^{23}
(c) 3.5×10^{23} (d) 2.5×10^{23}

16. The largest number of molecules is in

(a) $25g$ of CO_2
b) $46g$ of C_2H_5OH
(c) $36g$ of H_2O
d) $54g$ of N_2O_5

17. If 1 *M* and 2.5 *litre NaOH* solution is mixed with another 0.5 *M* and 3 *litre NaOH* solution, then molarity of the resultant solution will be

(a) 1.0 *M* (b) 0.73 *M*
(c) 0.80 *M* (d) 0.50 *M*

18. When a solute is present in trace quantities the following expression is used

(a) Gram per million
(b) Milligram percent
(c) Microgram percent
(d) Parts per million

19. When the concentration is expressed as the number of moles of a solute per litre of solution it known as

(a) Normality
(b) Molarity
(c) Mole fraction
(d) Mass percentage
(e) Molality

20. The normality of 2.3 *M* H_2SO_4 solution is

(a) 2.3 *N* (b) 4.6 *N*
(c) 0.46 *N* (d) 0.23 *N*